



Novel weak form quadrature element method with expanded Chebyshev nodes



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ABSTRACT

Based on the principle of minimum potential energy and the differential quadrature rule, novel weak form quadrature element method is proposed. Different from the existing ones, expanded Chebyshev grid points are used as the element nodes. A simple but general way is proposed to compute the strains at the integration points explicitly by using the differential quadrature rule. For illustration and verification, quadrature bar and beam elements are established. Several examples are given. Numerical results indicate that the proposed quadrature element method allows a longer time step as compared to elements with other nodes and is an accurate and efficient method for structural analysis.

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1. Introduction

The enhancement of computational efficiency and accuracy has always been a matter of interest to the research community of computational mechanics [1–15]. A variety of new and efficient methods have been proposed in recent years, such as the differential quadrature method (DQM) [1,2], discrete singular convolution (DSC) [3], harmonic differential quadrature (HDQ) [4,5], the Adomian decomposition method [6], the combined finite difference and differential transformation method [7], spectral finite element method [8,9], and the moving least square-Ritz method (MLS-Ritz) [10]. To simplify the non-linear formulations as well as to improve the efficiency of the DQM, the Hadamard and SJT products of matrices are introduced by Chen et al. [11–13]. The DSC and DQM have been successfully extended to solve problems of thin and thick plates with irregular shapes [14,15].

Based on the differential quadrature rule [1], the quadrature element method (QEM) has been proposed by Striz et al. [16]. Currently there are two kinds of quadrature element method. One, called the strong form QEM, is formulated based on the governing as well as boundary equations [16–18], and the other, called the weak form QEM, is formulated based on the principle of minimum potential energy [19–21]. It seems that the weak form QEM is more flexible than the strong form QEM and thus would find more applications in the area of structural mechanics, since the method is similar to the higher order finite element method and the time domain spectral element method.

The main character of the weak form QEM is that the differential quadrature rule is used in the formulation of the element stiffness matrix. In the formulations reported by Chen et al. [19], the element can be with any kind of nodes, such as uniformly distributed points, Gauss–Lobatto–Chebyshev points and Gauss quadrature points together with two end points. However, the derivations are complicated and thus the number of element nodes should be small, since it requires

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computing the inverse of a full Vandermonde matrix. To remove this deficiency, Gauss–Lobatto–Legendre (GLL) points are used as the element nodes and the GLL quadrature is used to obtain the element stiffness matrix and mass matrix [20,21]; thus the number of element nodes is not limited and the derivatives at nodal points can be calculated explicitly by the available formulations. This reduces the difficulty in calculations of the derivatives for high order elements, the round off error and programming effort. To use the differential quadrature rule, however, the element nodal points should be the same as the quadrature points; thus only GLL points can be used as the element nodes at the present time.

The expanded Chebyshev points are a very good approximation to the Lebesgue-optimal grid points, which allows the longest time step [22,23]. It is pointed out that the sensitivity to round off error, ease of programming, and maximum time step is likely to be far more important in choosing the “best” grid than slight variations in accuracy for dynamic problems [23].

The objective of this article is to propose a novel weak form quadrature element method. Different from the existing ones in [22,23], expanded Chebyshev grid points are adopted as element nodes. For the ease of programming and reducing the round off error, a simple but general way is proposed to compute the derivatives at the integration points explicitly by using the differential quadrature rule. For illustration and verifications, quadrature bar and beam elements are established. Several examples are given. Conclusions are drawn based on the results reported herein.

2. Weak form quadrature element formulations

The weak form quadrature element formulation process of the stiffness and mass matrices is analogous to the ones of the conventional finite element or time domain spectral element. The crucial ideas of the weak form QEM are adopting specific nodes and computing the derivatives at the integration points by using the differential quadrature rule. This may reduce the programming effort and the round off error, since explicit formulations are available to compute the derivatives at the nodal points.

2.1. Element nodes and shape functions

To achieve the longer time step, expanded Chebyshev points are adopted as the element nodes, namely [23],

$$\xi_k = -\frac{\cos((2k-1)\pi/(2N))}{\cos(\pi/(2N))} \quad k = 1, 2, \dots, N; \quad \xi \in [-1, 1]. \quad (1)$$

Depending on the problems, the shape functions can be either Lagrange interpolation functions or Hermit interpolation function. Without loss of generality, quadrature bar and beam elements are presented herein for simplicity.

2.2. Quadrature bar element

For a weak form N -node quadrature bar element, the displacement within the element is assumed as

$$u(\xi, t) = \sum_{j=1}^N l_j(\xi) u(\xi_j, t) = \sum_{j=1}^N l_j(\xi) u_j \quad (2)$$

where $u(\xi, t)$ is the axial displacement, t is the time, and the shape functions are Lagrange interpolation functions defined by

$$l_j(\xi) = \frac{(\xi - \xi_1)(\xi - \xi_2) \cdots (\xi - \xi_{j-1})(\xi - \xi_{j+1}) \cdots (\xi - \xi_N)}{(\xi_j - \xi_1)(\xi_j - \xi_2) \cdots (\xi_j - \xi_{j-1})(\xi_j - \xi_{j+1}) \cdots (\xi_j - \xi_N)} = \prod_{\substack{k=1 \\ k \neq j}}^N \frac{\xi - \xi_k}{\xi_j - \xi_k}. \quad (3)$$

Substituting Eq. (2) into the strain energy and kinematic energy expressions of a bar element results

$$\begin{aligned} U &= \frac{1}{2} \int_0^L EA \left(\sum_{j=1}^N \frac{dl_j(x)}{dx} u_j(t) \right)^2 dx = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \left(\sum_{j=1}^N l'_j(\xi) u_j(t) \right)^2 d\xi \\ &= \frac{1}{2} \{u\}^T [k] \{u\} \end{aligned} \quad (4)$$

$$\begin{aligned} T &= \frac{1}{2} \int_0^L \rho A \left(\sum_{j=1}^N l_j(x) \dot{u}_j(t) \right)^2 dx = \frac{1}{2} \int_{-1}^1 \frac{\rho AL}{2} \left(\sum_{j=1}^N l_j(\xi) \dot{u}_j(t) \right)^2 d\xi \\ &= \frac{1}{2} \{\dot{u}\}^T [m] \{\dot{u}\} \end{aligned} \quad (5)$$

where E , A , L and ρ are the elasticity modulus, cross sectional area, element length and mass density of the bar material, $[k]$ and $[m]$ are the stiffness matrix and mass matrix, $\{u\}$ and $\{\dot{u}\}$ are nodal displacement vector and nodal velocity vector,

respectively, the prime is the first order derivative with respect to ξ , and the over dot denotes the first order derivative with respect to time t .

Similar to the conventional finite element method, the stiffness matrix and mass matrix can be obtained by numerical integration, such as by the method of Gauss quadrature or GLL quadrature. If GLL points are used as the element nodes and the GLL quadrature is used in the integration, then

$$k_{ij} = \frac{2EA}{L} \sum_{k=1}^N \bar{H}_k l'_i(\xi_k) l'_j(\xi_k) = \frac{2EA}{L} \sum_{k=1}^N \bar{H}_k A_{ki} A_{kj} \quad (i, j = 1, 2, \dots, N) \quad (6)$$

$$m_{ij} = \frac{\rho AL}{2} \sum_{k=1}^N \bar{H}_k l_i(\xi_k) l_j(\xi_k) = \frac{\rho AL}{2} \sum_{k=1}^N \bar{H}_k \delta_{ki} \delta_{kj} \quad (i, j = 1, 2, \dots, N) \quad (7)$$

where $\bar{H}_k$ are the weights in the GLL quadrature, A_{ij} is the weighting coefficient of the first order derivative with respect to ξ , computed explicitly by [2]

$$A_{ij} = l'_j(\xi_i) = \begin{cases} \frac{\prod_{k=1, k \neq i, j}^N (\xi_i - \xi_k)}{\prod_{k=1, k \neq j}^N (\xi_j - \xi_k)} & (i \neq j) \\ \sum_{k=1, k \neq i}^N \frac{1}{(\xi_i - \xi_k)} & (i = j). \end{cases} \quad (8)$$

It is seen that the formulation of the weak form quadrature element is very simple due to using the differential quadrature rule to compute the first order derivative of the displacement at the integration points. Besides, the mass matrix is a diagonal matrix. $[k]$ and $[m]$ of the N -node bar element are exactly the same as the ones of the time domain spectral element [8] if the same nodes and quadrature rule are used.

For the expanded Chebyshev nodes, however, Eqs. (6) and (7) are no longer valid, since the nodes are not the same as the integration points. This is perhaps one of the main reasons why only GLL points are used as the element nodes at the present time, although GLL points are not the Lebesgue-optimal grid points.

To solve the difficulty in the formulation of the weak form quadrature element with nodes other than the GLL points, a simple but general method is proposed to compute the derivative of the displacement at integration points by using the differential quadrature rule. For illustration, the Gauss quadrature is used, since it is slightly more accurate than the GLL quadrature.

Denote the Gauss quadrature points as ς_k ($k = 1, 2, \dots, N$) and the corresponding weights as H_k ($k = 1, 2, \dots, N$). For the shape function $l_j(\xi)$, the function values at the N Gauss quadrature points can be calculated by

$$l_j(\xi)_{\xi=\varsigma_k} = l_j(\varsigma_k) = l_{jk}. \quad (9)$$

Since the shape function of the N -node bar element is an $(N - 1)$ th order polynomial, it can be interpolated by

$$l_j(\xi) = \sum_{k=1}^N \tilde{l}_k(\xi) l_j(\varsigma_k) = \sum_{k=1}^N \tilde{l}_k(\xi) l_{jk} \quad (10)$$

where $\tilde{l}_k(\xi)$ are Lagrange interpolation functions based on the N Gauss quadrature points ς_k ($k = 1, 2, \dots, N$), which can be calculated by

$$\tilde{l}_k(\xi) = \frac{(\xi - \varsigma_1)(\xi - \varsigma_2) \cdots (\xi - \varsigma_{k-1})(\xi - \varsigma_{k+1}) \cdots (\xi - \varsigma_N)}{(\varsigma_k - \varsigma_1)(\varsigma_k - \varsigma_2) \cdots (\varsigma_k - \varsigma_{k-1})(\varsigma_k - \varsigma_{k+1}) \cdots (\varsigma_k - \varsigma_N)} = \prod_{\substack{m=1 \\ m \neq k}}^N \frac{\xi - \varsigma_m}{\varsigma_k - \varsigma_m}. \quad (11)$$

It is seen that Eq. (11) is different from Eq. (3) although the form is similar. Besides, Eq. (10) is exact since both sides are $(N - 1)$ th order polynomials.

Therefore, the weighting coefficients of the first order derivative of the shape function at the integration points can be computed by using the differential quadrature rule, namely,

$$\bar{A}_{ij} = l'_j(\varsigma_i) = \sum_{k=1}^N \tilde{l}'_k(\varsigma_i) l_j(\varsigma_k) = \sum_{k=1}^N \tilde{l}'_k(\varsigma_i) l_{jk} = \sum_{k=1}^N \bar{A}_{ik} l_{jk} \quad (12)$$

where the over-bar means that the weighting coefficients are computed by Eq. (8) but based on the integration points and not on the element nodes. In other words, $\bar{A}_{ij}$ are computed by

$$\bar{A}_{ij} = \bar{l}_j(\xi_i) = \begin{cases} \prod_{\substack{k=1 \\ k \neq i, j}}^N (\xi_i - \xi_k) / \prod_{\substack{k=1 \\ k \neq j}}^N (\xi_j - \xi_k) & (i \neq j) \\ \sum_{\substack{k=1 \\ k \neq i}}^N \frac{1}{(\xi_i - \xi_k)} & (i = j). \end{cases} \quad (13)$$

If the element nodal points are the same as the integration points, such as in the case of GLL nodes and GLL quadrature, $\xi_i = \xi_i$, $l_{jk} = \delta_{jk}$. Thus one has $A_{ij} = \bar{A}_{ij}$.

Once $\bar{A}_{ij}$ are known, the elements of stiffness matrix can be computed by

$$k_{ij} = \frac{2EA}{L} \sum_{k=1}^N H_k l'_i(\xi_k) l'_j(\xi_k) = \frac{2EA}{L} \sum_{k=1}^N H_k \bar{A}_{ki} \bar{A}_{kj} \quad (i, j = 1, 2, \dots, N). \quad (14)$$

To obtain a diagonal mass matrix, the technique of row summation is adopted [9]. Thus, the diagonal term in the mass matrix can be computed by

$$m_{ij} = \frac{\rho AL}{2} \sum_{k=1}^N H_k l_j(\xi_k) = \frac{\rho AL}{2} \sum_{k=1}^N H_k l_{jk} \quad (j = 1, 2, \dots, N). \quad (15)$$

It is seen that the proposed method is simple and can be used for any kind of nodes and quadrature rule. Since the explicit formula for $u'(\xi, t)$ is not needed, the effort for deriving an N -node element is greatly reduced.

2.3. Quadrature beam element with Hermit interpolation

For a weak form N -node quadrature beam element, the displacement within the element can be assumed as

$$w(\xi, t) = \sum_{j=1}^N \varphi_j(\xi) w(\xi_j, t) + \psi_1(\xi) w'(\xi_1, t) + \psi_N(\xi) w'(\xi_N, t) = \sum_{j=1}^{N+2} h_j(\xi) \delta_j \quad (16)$$

where w is the deflection, t is the time, $\{\delta\} = [w_1, w'_1, w_2, w_3, \dots, w_{N-1}, w_N, w'_N]^T$, the prime is the first order derivative, and the shape functions are Hermit interpolation functions defined by [18]

$$\begin{aligned} \psi_j(\xi) &= \frac{1}{\xi_j - \xi_{N-j+1}} l_j(\xi) (\xi - \xi_j) (\xi - \xi_{N-j+1}) \quad (j = 1, N) \\ \varphi_j(\xi) &= \begin{cases} \frac{1}{\xi_j - \xi_{N-j+1}} l_j(\xi) (\xi - \xi_{N-j+1}) - \left[l'_j(\xi_j) + \frac{1}{\xi_j - \xi_{N-j+1}} \right] \psi_j(\xi) & (j = 1, N) \\ \frac{1}{(\xi_j - \xi_1)(\xi_j - \xi_N)} l_j(\xi) (\xi - \xi_1) (\xi - \xi_N) & (j = 2, 3, \dots, N-1). \end{cases} \end{aligned} \quad (17)$$

In Eq. (16), $l_j(\xi)$ are Lagrange interpolation functions defined by Eq. (3). The prime is the first order derivative with respect to ξ .

Substituting Eq. (16) into the strain energy and kinematic energy expressions of a beam element results

$$U = \frac{1}{2} \int_{-1}^1 \frac{8EI}{L^3} \left(\sum_{j=1}^{N+2} h'_j(\xi) \delta_j(t) \right)^2 d\xi = \frac{1}{2} \{\delta\}^T [k] \{\delta\} \quad (18)$$

$$T = \frac{1}{2} \int_{-1}^1 \frac{\rho AL}{2} \left(\sum_{j=1}^N l_j(x) \dot{w}_j(t) \right)^2 d\xi = \frac{1}{2} \{\dot{w}\}^T [m] \{\dot{w}\} \quad (19)$$

where E, I, A, L and ρ are the elasticity modulus, moment of inertia, cross sectional area, element length and mass density of the beam material, $[k]$ and $[m]$ are the stiffness matrix and mass matrix, $\{\delta\}$ and $\{\dot{w}\}$ are nodal displacement vector and nodal velocity vector, respectively, the double prime is the second order derivative with respect to ξ , and the over dot denotes the first order derivative with respect to time t .

Note that different displacement functions are used to formulate the element stiffness matrix and the mass matrix for simplicity. Since Eq. (19) is exactly the same as Eq. (5), only the formulation of stiffness matrix is focused on in what follows.

If GLL points are used as the element nodes and the GLL quadrature is used in the integration, then

$$k_{ij} = \frac{8EI}{L^3} \sum_{k=1}^N \bar{H}_k h_i''(\xi_k) h_j''(\xi_k) = \frac{8EI}{L^3} \sum_{k=1}^N \bar{H}_k B_{ki} B_{kj} \quad (i, j = 1, 2, \dots, N+2) \quad (20)$$

where B_{ki} and B_{kj} are the weighting coefficients of the second order derivative with respect to ξ . The explicit formula is available [18] for computing B_{ki} and B_{kj} . Since the explicit formula is far more complicated than Eq. (8) and not used herein, the formula is omitted. Interested readers may refer to [18] for details. For the expanded Chebyshev nodes with the Gauss quadrature, however, B_{ki} and B_{kj} should be computed differently.

It is known that the shape function in Eq. (16) is an $(N+1)$ th order polynomial, which can be interpolated by

$$h_j(\xi) = \sum_{k=1}^{N+2} \hat{l}_k(\xi) h_j(\varsigma_k) = \sum_{k=1}^{N+2} \hat{l}_k(\xi) h_{jk} \quad (21)$$

where $\hat{l}_k(\xi)$ are Lagrange interpolation functions based on the $N+2$ Gauss quadrature points ς_k ($k = 1, 2, \dots, N+2$). $\hat{l}_k(\xi)$ can be calculated by

$$\hat{l}_k(\xi) = \frac{(\xi - \varsigma_1)(\xi - \varsigma_2) \cdots (\xi - \varsigma_{k-1})(\xi - \varsigma_{k+1}) \cdots (\xi - \varsigma_{N+2})}{(\varsigma_k - \varsigma_1)(\varsigma_k - \varsigma_2) \cdots (\varsigma_k - \varsigma_{k-1})(\varsigma_k - \varsigma_{k+1}) \cdots (\varsigma_k - \varsigma_{N+2})} = \prod_{\substack{m=1 \\ m \neq k}}^{N+2} \frac{\xi - \varsigma_m}{\varsigma_k - \varsigma_m}. \quad (22)$$

h_{jk} in Eq. (21) are the values of shape function at the Gauss quadrature points, namely,

$$h_j(\xi)_{\xi=\varsigma_k} = h_j(\varsigma_k) = h_{jk}. \quad (23)$$

Similar to Eq. (10), Eq. (21) is exact since both sides are $(N+1)$ th order polynomials.

With the proposed method, the weighting coefficients of the second order derivative of the shape function at the integration points can be computed by

$$\bar{B}_{ij} = h_j''(\varsigma_i) = \sum_{k=1}^{N+2} \hat{l}_k'(\varsigma_i) h_j(\varsigma_k) = \sum_{k=1}^{N+2} \hat{l}_k'(\varsigma_i) h_{jk} = \sum_{k=1}^{N+2} \hat{B}_{ik} h_{jk} \quad (24)$$

where the over-bar means that the weighting coefficients are computed based on the integration points and not on the element nodes. $\hat{B}_{ik}$ can be conveniently computed by

$$\hat{B}_{ik} = \sum_{l=1}^{N+2} \hat{A}_{il} \hat{A}_{lk} \quad (25)$$

in which $\hat{A}_{ij}$ can be explicitly computed by

$$\hat{A}_{ij} = \hat{l}_j'(\varsigma_i) = \begin{cases} \frac{\prod_{\substack{k=1 \\ k \neq i, j}}^{N+2} (\varsigma_i - \varsigma_k)}{\prod_{\substack{k=1 \\ k \neq j}}^{N+2} (\varsigma_j - \varsigma_k)} & (i \neq j) \\ \sum_{\substack{k=1 \\ k \neq i}}^{N+2} \frac{1}{(\varsigma_i - \varsigma_k)} & (i = j). \end{cases} \quad (26)$$

It is seen that the way to compute the weighting coefficients by the proposed method is far simpler than the explicit formula provided in [18]. Once $\bar{B}_{ij}$ are known, the elements in stiffness matrix can be computed by

$$k_{ij} = \frac{8EI}{L^3} \sum_{k=1}^{N+2} H_k h_i''(\varsigma_k) h_j''(\varsigma_k) = \frac{8EI}{L^3} \sum_{k=1}^{N+2} H_k \bar{B}_{ki} \bar{B}_{kj} \quad (i, j = 1, 2, \dots, N+2). \quad (27)$$

It is known that the element stiffness matrix can be exactly integrated by using the Gauss quadrature with N points. To reduce the number of integration points, ς_k ($k = 1, 2, \dots, N+2$) can be taken as $-1, N$ Gauss quadrature points, $+1$ for computing $\bar{B}_{ij}$ by the proposed method. Then, Eq. (27) is modified to

$$k_{ij} = \frac{8EI}{L^3} \sum_{k=2}^{N+1} H_k \bar{B}_{ki} \bar{B}_{kj} \quad (i, j = 1, 2, \dots, N+2). \quad (28)$$

In this way, the stiffness matrix is integrated by using the Gauss quadrature with N points. If reduced integration is needed, ς_k should be modified in a similar way such that the required Gauss quadrature points are contained. The remaining points seem unimportant but are better selected within $[-1, 1]$ and ranged in an increasing order. The summation range in Eq. (28) should be changed accordingly for the reduced integration.

2.4. Quadrature beam element with Lagrange interpolation

Instead of using Eq. (16), w can also be assumed by using Lagrange interpolation functions, namely,

$$w(\xi, t) = \sum_{j=1}^N l_j(\xi) w(\xi_j, t) = \sum_{j=1}^N l_j(\xi) w_j \quad (29)$$

where $l_j(\xi)$ are Lagrange interpolation functions and defined by Eq. (3).

To introduce the derivative degrees of freedom at two end points, the built-in-method-3 [24] can be used. In other words, w'_1, w'_N are introduced during computing $\bar{B}_{ij}$. Let $\{\bar{\delta}\} = [w_1, w_2, w_3, \dots, w_{N-1}, w_N, w'_1, w'_N]^T$ for convenience in presentation; then

$$\begin{aligned} \bar{B}_{1k} &= \sum_{l=2}^{N-1} \bar{A}_{il} A_{lk}; & \bar{B}_{1(N+1)} &= \bar{A}_{11}; & \bar{B}_{1(N+2)} &= \bar{A}_{1N} \\ \bar{B}_{ik} &= \sum_{l=1}^N \bar{A}_{il} A_{lk}; & \bar{B}_{i(N+1)} &= \bar{B}_{i(N+2)} = 0 & (i = 2, 3, \dots, N-1) \\ \bar{B}_{Nk} &= \sum_{l=2}^{N-1} \bar{A}_{il} A_{lk}; & \bar{B}_{N(N+1)} &= \bar{A}_{N1}; & \bar{B}_{N(N+2)} &= \bar{A}_{NN} \quad (k = 1, 2, \dots, N) \end{aligned} \quad (30)$$

where $\bar{A}_{il}$ and A_{lk} are weighting coefficients of the first order derivatives computed by Eqs. (12) and (8). The difference between $\bar{A}_{il}$ and A_{lk} is that $\bar{A}_{il}$ are computed based on the integration points but A_{lk} are computed based on the node points. Rearrangement may be necessary if the displacement vector is arranged as $\{\delta\} = [w_1, w'_1, w_2, w_3, \dots, w_{N-1}, w_N, w'_N]^T$. The elements of stiffness matrix are computed by

$$k_{ij} = \frac{8EI}{L^3} \sum_{k=1}^N H_k \bar{B}_{ki} \bar{B}_{kj} \quad (i, j = 1, 2, \dots, N+2). \quad (31)$$

The mass matrix remains the same. It is seen that the formulation of a weak form quadrature beam element is simple. The research achievements in the differential quadrature method may be directly used.

3. Numerical results and discussion

To demonstrate the accuracy and efficiency of the quadrature element method, numerical results are given. Three different kinds of nodes, i.e., expanded Chebyshev nodes, GLL nodes and uniformly distributed nodes, are investigated and denoted by symbols “Expanded”, “GLL” and “Uniform”, respectively.

Example 1. Find the nature frequencies for a cantilever bar vibrating freely in the axial direction by using one N -node quadrature bar element.

Similar to the displacement finite element method, only essential boundary conditions need to be applied. After applying $u_1 = 0$, an $(N-1) \times (N-1)$ matrix equation is directly solved by using a generalized eigenvalue solver. Thus, $(N-1)$ frequencies are obtained.

The number of nodes varies from 5 to 9 for the convergence study. The frequency parameters, defined by $\lambda = (\omega L / \pi) \sqrt{\rho / E}$, are listed in Table 1. The exact solutions are cited from Ref. [25].

It is seen that very accurate fundamental frequency can be obtained with one 5-node quadrature bar element for the case of cantilever bar. The accuracy of higher mode frequencies increases with the increase of number of nodal points. Among the three sets of nodes, the expanded Chebyshev nodes and GLL nodes yield similar solution accuracy for the first five mode frequencies. The element with uniformly distributed nodes yields the least accurate solutions. It should be pointed out that, however, the solution accuracy of the quadrature element may also depend on the boundary conditions.

Example 2. Find the nature frequencies for a free-free beam vibrating freely in the transverse direction by using one N -node quadrature beam element.

Since only essential boundary conditions need to be applied, no boundary conditions are specified for the free-free beam. An $(N+2) \times (N+2)$ matrix equation is directly solved by using a generalized eigenvalue solver to obtain the frequencies. Note that only $(N-2)$ non-zero frequencies are obtained, since the terms related to the two derivative degrees of freedom in the mass matrix are zero.

Three different kinds of nodes, i.e., uniformly distributed nodes, GLL nodes and expanded Chebyshev nodes, are investigated. The number of nodes varies from 5 to 9 for the convergence study. As is expected, the stiffness matrix is exactly the same whether (N) point or $(N+2)$ point Gauss quadrature is used.

Table 1Convergence of frequency parameters of the CF bar by one quadrature bar element ($\lambda = (\omega L/\pi)\sqrt{\rho/E}$).

N	Node types	Mode number				
		1	2	3	4	5
5	Expanded	0.5000	1.5041	2.5295	3.8746	–
	GLL	0.5000	1.4983	2.4308	4.2356	–
	Uniform	0.5000	1.4878	2.7104	4.9963	–
6	Expanded	0.5000	1.5003	2.5261	3.6162	5.4983
	GLL	0.5000	1.4999	2.4852	3.4602	6.2554
	Uniform	0.4999	1.4829	2.6366	3.8705	7.6323
7	Expanded	0.5000	1.4998	2.5020	3.5790	4.8057
	GLL	0.5000	1.5000	2.4987	3.4540	4.6530
	Uniform	0.5000	1.4779	2.2299	4.1153	10.144
8	Expanded	0.5000	1.5000	2.4997	3.5111	4.6703
	GLL	0.5000	1.5000	2.4999	3.4917	4.4370
	Uniform	0.4999	1.4879	2.4116	3.5717	4.6669
9	Expanded	0.5000	1.5000	2.4998	3.5008	4.6319
	GLL	0.5000	1.5000	2.5000	3.4992	4.4715
	Uniform	0.5000	1.4809	2.1625	2.8012	5.6775
	Exact [25]	0.5000	1.5000	2.5000	3.5000	4.5000

Table 2Convergence of frequency parameters for the FF beam by one quadrature beam element with the Hermit interpolation ($\lambda = \omega L^2 \sqrt{\rho A/EI}$).

N	Node types	Mode number				
		1	2	3	4	5
5	Expanded	22.5722	66.3258	89.8472	–	–
	GLL	22.3056	76.6811	116.681	–	–
	Uniform	24.0499	48.6505	113.647	–	–
6	Expanded	22.4591	59.9582	143.198	162.642	–
	GLL	22.3868	60.6278	174.387	226.508	–
	Uniform	23.2451	56.0030	99.7904	196.313	–
7	Expanded	22.3518	62.5062	113.080	267.101	278.377
	GLL	22.3727	61.9972	116.524	337.989	402.973
	Uniform	22.0497	75.1933	97.9405	299.640	574.328
8	Expanded	22.3663	61.8384	122.402	178.602	450.431
	GLL	22.3733	61.6485	123.150	189.820	593.206
	Uniform	22.2126	68.5747	114.518	167.312	342.992
9	Expanded	22.3741	61.5477	121.559	205.256	258.049
	GLL	22.3733	61.6750	120.639	208.613	283.160
	Uniform	22.3295	55.5929	165.590	168.629	–
	Exact [26]	22.373	61.673	120.903	199.859	298.556

The frequency parameters, defined by $\lambda = \omega L^2 \sqrt{\rho A/EI}$, are summarized in Tables 2 and 3. The results shown in Table 2 are obtained by using one quadrature beam element with Hermit interpolations, and the results shown in Table 3 are obtained by using one quadrature beam element with Lagrange interpolations. The exact solutions are cited from Ref. [26].

It is seen that accurate fundamental frequency can be obtained with one 8-node quadrature beam element. The accuracy of higher mode frequencies increases with the increase of number of nodal points. Among the three sets of nodes, the GLL nodes yield the most accurate solutions and the uniformly distributed nodes yield the least accurate solutions. Due to neglecting the terms in the mass matrix related to derivative degrees, the solution accuracy of beam element with expanded Chebyshev nodes is not as good as the case of bar element. Overall speaking, however, the solution accuracy by the expanded Chebyshev nodes and GLL nodes are similar. Since some terms in the diagonal mass matrix are negative when $N > 8$, only four results are listed in Tables 2 and 3 for the 9-node element with uniformly distributed nodes. Similar phenomenon is reported in [14,27] for spectral plate elements.

To investigate the effect of quadrature rules on the solution accuracy of beam element with GLL nodes, both the Gauss quadrature and the GLL quadrature, denoted by GLL(1) and GLL(2) in Table 3, are used to integrate the stiffness matrix. It is seen that the GLL quadrature yields slightly better results although the stiffness matrix is not fully integrated. However, the difference is small.

The length of the maximum stable time step is an important issue in solving time-dependent partial differential equations. The expanded Chebyshev nodes allow a longer time step because the clustering of points near the boundary is not as extreme as for the GLL nodes [23]. It is reported that the allowed maximum time step, Δt_{cr} , is inversely proportional to $\omega_{\max}^e$, i.e., $\Delta t_{cr} < 2/\omega_{\max}^e$ [27], where superscript e denotes the element.

Thus the maximum frequency $\omega_{\max}^e$ of the quadrature element is computed and compared. $\omega_{\max}^e$ is obtained by solving the following equation,

$$[k]^e - \omega^2 [M]^e = 0. \quad (32)$$

Table 3Convergence of frequency parameters for the FF beam by one quadrature beam element with Lagrange interpolation ($\lambda = \omega L^2 \sqrt{\rho A/EI}$).

N	Node types	Mode number				
		1	2	3	4	5
5	Expanded	22.9004	52.3224	52.4373	–	–
	GLL(1)	22.3943	53.8390	59.1372	–	–
	GLL(2)	22.0578	76.6812	96.3403	–	–
	Uniform	23.4187	41.3874	77.0381	–	–
6	Expanded	22.5445	62.9539	84.4046	101.505	–
	GLL(1)	22.4256	59.8706	92.0722	117.770	–
	GLL(2)	22.3868	58.6874	174.387	192.031	–
	Uniform	23.3840	54.0487	84.4761	138.720	–
7	Expanded	22.3356	63.1206	115.776	138.576	171.666
	GLL(1)	22.3645	62.1351	105.054	159.126	202.216
	GLL(2)	22.3725	61.9972	110.293	337.989	353.009
	Uniform	21.8281	73.4623	97.6241	275.057	428.840
8	Expanded	22.3642	61.6015	124.042	165.500	235.340
	GLL(1)	22.3719	61.5054	122.780	153.463	267.362
	GLL(2)	22.3733	61.6338	123.150	176.791	593.206
	Uniform	22.3356	63.1206	115.776	138.576	171.666
9	Expanded	22.3721	61.5059	120.166	207.098	224.218
	GLL(1)	22.3722	61.6479	119.477	205.291	210.762
	GLL(2)	22.3733	61.6750	120.455	208.613	261.469
	Uniform	22.2247	55.2709	171.470	171.740	–
	Exact [26]	22.373	61.673	120.903	199.859	298.556

Table 4The highest frequency parameter of the quadrature bar element ($\lambda_{\max}^e = \omega_{\max}^e L \sqrt{\rho/E}$).

Node types	N		
	5	6	7
Expanded	12.6051 (1.000)	17.7004 (1.000)	23.9879 (1.000)
GLL	13.5406 (1.074)	19.7980 (1.119)	27.3780 (1.141)
Uniform	16.0462 (1.273)	24.3363 (1.375)	67.4197 (2.811)
QEM-55	12.5664 (4)	15.7080 (5)	18.8496 (6)

Table 5The highest frequency parameter of the quadrature beam element with Hermit interpolation ($\lambda_{\max}^e = \omega_{\max}^e L^2 \sqrt{\rho A/EI}$).

Node types	N		
	5	6	7
Expanded	89.8742 (1.000)	162.642 (1.000)	278.377 (1.000)
GLL(1)	96.3403 (1.101)	186.142 (1.172)	337.989 (1.255)
GLL(2)	116.681 (1.298)	226.508 (1.393)	402.973 (1.448)
Uniform	113.647 (1.265)	196.313 (1.207)	574.328 (2.063)
Exact [26]	120.903 (3)	199.859 (4)	298.556 (5)

Table 6The highest frequency parameter of the quadrature beam element with Lagrange interpolation ($\lambda_{\max}^e = \omega_{\max}^e L^2 \sqrt{\rho A/EI}$).

Node types	N		
	5	6	7
Expanded	52.4373 (1.000)	101.505 (1.000)	171.666 (1.000)
GLL(1)	59.1372 (1.128)	117.770 (1.160)	202.216 (1.178)
GLL(2)	96.3403 (1.837)	192.031 (1.892)	353.009 (2.056)
Uniform	77.0381 (1.469)	138.720 (1.367)	428.840 (2.498)
Exact [26]	120.903 (3)	199.859 (4)	298.556 (5)

The dimension of the matrices $[k]^e$ and $[M]^e$ is $N \times N$ and $(N + 2) \times (N + 2)$ for the N -node bar and beam elements, respectively.

The maximum frequency parameters obtained by the quadrature bar element, beam element with Hermit interpolations and beam element with Lagrange interpolations are summarized in Tables 4–6, respectively. It is confirmed that the quadrature elements with expanded Chebyshev nodes allow a longer time step as compared to other elements, since $\omega_{\max}^e$ is the smallest one. The allowed maximum time step is the smallest for elements with uniformly distributed nodes. The difference increases with the increase of number of elemental nodes.

The accurate solutions are included in Tables 4–6. In Table 4, QEM-55 denotes that the solution is obtained by using a 55-node element. Thus the results should be very accurate; increasing the node number to 105, the same results are obtained. The number appeared in parentheses is the mode number. From Table 4, it is seen that the highest frequency is always greater than the accurate solution. On the other hand, the highest frequency listed in Tables 5 and 6 is either lower or greater than the accurate solution depending on the node number. This is perhaps due to the approximation of the mass matrix and needs to be investigated further.

4. Conclusions

The novel weak form quadrature element method is proposed. Different from the existing ones, expanded Chebyshev grid points are used as the element nodes. To use the differential quadrature rule for calculations of the derivatives at the integration points, a simple but general way is proposed. It is seen that the formulation of a weak form quadrature element is simple with the differential quadrature rule. For illustration and verifications, quadrature bar and beam elements are established. Numerical results indicate that the element with expanded Chebyshev nodes allows a longer time step and that the proposed quadrature element method is an accurate and efficient method for structural analysis. The proposed method can be extended to two-dimensional quadrilateral plate element if the tensor product of one-dimensional nodes is used as the nodal points of the plate element. The research is underway.

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